Sijill — Software Requirements Specification (SRS)

Nov 8, 2026 · @Abdalwahab · Version 1.0 (Phase 1 baseline)

1. Introduction

This SRS defines what Phase 1 of Sijill (Sijill Files) must do so the build can start and be tested against clear requirements. It follows the IEEE 830 structure and is derived from the Sijill System Proposal.

1.1 Purpose

The document is for the development team, the senior reviewer, the network's leadership (sponsor) and each school's IT admin. It is the baseline for design, testing and acceptance of Phase 1. Phases 2–4 are summarised in the Appendix and will get their own SRS before their build starts.

1.2 Scope

Sijill ("the register") is a read-only checker for the school files of a network of five schools and a head office. It reads each file's location, type and sharing, applies configurable rules, tells the right person what is wrong and how to fix it, and follows up until a re-check shows the file is fixed.

- **In scope (Phase 1):** read-only connection per school domain; file reading and classification; 10 starter rules; issue lifecycle; bilingual messages and reminders; five role views; per-file history and tamper-evident activity log; leadership alerts for major incidents; independent in-country backup; planning-file review; name-free exports; deletion protection; Arabic-first interface.
- **Out of scope:** any write by Sijill to school files; replacing the workspace, school email or QES; physical servers.

1.3 Definitions

Term	Meaning
Issue	One rule broken by one file

Term	Meaning
Rule	Versioned configuration: check type, scope, severity, reminder days, Arabic and English fix steps
Check type	One of 9 kinds of test a rule can apply (location, public link, outside sharing, etc.)
Re-check	A fresh read of one file; the only way an issue closes as Fixed
Exception	A coordinator-approved pause of a rule for one file, with a reason and an end date
Major incident	Exam leak (R6), important deletion (R10), 20+ deletions in an hour, or an urgent issue on Restricted student or health data
Sensitivity	Public, Internal, Confidential or Restricted
Workspace	The school's cloud office suite (drives, directory, email)
QES	The Ministry's education system used by government schools
School day	Sunday–Thursday, 07:00–16:00

1.4 References

- Sijill — System Proposal (29 pages), sections 1–15
- Sijill — Associate Brief v3 and product prototype
- Qatar Personal Data Privacy Protection Law (Law No. 13 of 2016)
- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications

1.5 Conventions

Requirement IDs follow the proposal: FR-x.y for functional, NFR-x for non-functional, SEC-x for security, DR-x for data. Priority is **Must** (needed for the Phase 1 gate), **Should** (Phase 1, can slip to the last stage) or **Could** (nice to have).

2. Overall description

Sijill sits beside each school's existing cloud workspace: it reads files through read-only access and sends out only messages and encrypted backup copies. Staff keep working in the workspace as they do today.

2.1 Product perspective

Sijill is a new, self-contained web system built as a Spring Boot modular monolith on one PostgreSQL database. It connects to one workspace domain per school through a read-only service account, to an in-country AI service for document understanding, to an email sender for messages, and to a second in-country cloud provider for backup. It does not replace QES; attendance, timetable, groups and live lessons stay there.

2.2 Product functions

1. **Check:** full check of every shared and personal drive nightly at 02:00, plus a change feed within minutes.
2. **Understand:** read each file in memory, mask personal data, classify document type and sensitivity with evidence.
3. **Apply rules:** test each file against active rules; each broken rule becomes one issue.
4. **Tell and follow up:** send the owner and coordinator a bilingual message (what, why, how to fix), then reminders and follow-ups.
5. **Close only on re-check:** "I've fixed it" triggers a re-check; the issue closes only if the file now passes.
6. **Oversee:** school health score, principal and leadership views, major-incident alerts, weekly report.
7. **Record:** per-file history and an append-only, hash-chained activity log.
8. **Protect:** hourly backup to a second provider, deletion alerts, restore requests.
9. **Review plans:** teachers submit weekly planning files; coordinators approve or return them.

2.3 User classes

Role	Count	Main tasks	Sees
Teacher	~100	Fix own issues, submit planning files	Own files and issues only

Role	Count	Main tasks	Sees
Subject coordinator	1 per subject per school	Fix sharing, classify unclear files, decide disputes, allow exceptions, review plans	Her subject's files and issues
Principal	5	Read-only oversight	All of her school's notifications, score, issues, log
School IT admin	5	Connection, people sync, rules, settings	Her school's setup; counts, not file content
Leadership	Network executive team	Major incidents, network totals, weekly report, approve network rules	All schools as totals

All roles come from groups in the school's workspace directory. Total: about 124 staff across 6 domains.

2.4 Operating environment

- Server: Java 21, Spring Boot, PostgreSQL (row-level security, pgvector), message queue, Docker containers, hosted in-country on the cloud.
- Client: current Chrome, Edge, Safari and Firefox on phone, tablet and laptop; Arabic right-to-left first, English optional.
- External: workspace APIs (files, permissions, directory, audit feed), in-country AI model, email sender account, backup provider (Provider B).

2.5 Design and implementation constraints

- Sijill must never move, edit, share or delete a school file; read-only scopes make a write impossible.
- All data, logs, AI processing and backups stay in Qatar.
- File content is read in memory and discarded; only metadata, masked evidence and profiles are stored.
- Rules, roles, sites, document types, languages and timings are configuration, not code.
- Only synthetic data is used until a data processing agreement is signed.
- Scores belong to schools, never to individuals.

2.6 Assumptions and dependencies

- Each school has its own workspace domain whose plan allows read-only, domain-wide access to drives, directory and audit feed.
- Each subject coordinator manages her subject's shared drive.
- Each school can name an academic staff member as School IT admin.
- Directory groups can be tidied to map to the five roles.
- An AI service and a second cloud provider are available in-country.

3. Functional requirements

Phase 1 has 12 modules and 72 requirements; 64 are Must and 8 are Should. Each requirement is testable on its own, and new ones are added as new rows.

3.1 Checking and understanding (FR-1)

ID	Requirement	Priority
FR-1.1	The system shall run a full check of all shared and personal drives of every connected school nightly at 02:00.	Must
FR-1.2	The system shall process the workspace change feed and check changed files within 10 minutes.	Must
FR-1.3	The system shall read file text in memory (including Arabic and English OCR for scans) and discard it after classification.	Must
FR-1.4	The system shall mask names, IDs, +974 phone numbers and health words before any AI call.	Must
FR-1.5	The system shall store a file profile: document type, site, subject, student/staff data flag, sensitivity, confidence, masked evidence, model version.	Must
FR-1.6	Hard checks (ID numbers, phone numbers, student IDs, health words) shall only raise sensitivity, never lower it.	Must

ID	Requirement	Priority
FR-1.7	Files below the confidence threshold, or unreadable, shall go to Needs review for the coordinator, never reported as fine.	Must
FR-1.8	Locked, oversized or disallowed file types shall be skipped and listed in Recent checks.	Must

3.2 Rules (FR-2)

ID	Requirement	Priority
FR-2.1	The system shall ship 10 starter rules (R1–R10) as data, each combining one of 9 check types with document types, sensitivity, severity and reminder days.	Must
FR-2.2	Editing a rule shall create a new version; each issue keeps the rule version that raised it.	Must
FR-2.3	The rule editor shall show a live impact preview (dry run on the last check: "about N files would break this rule") and a live message preview.	Must
FR-2.4	A rule can be saved as draft, saved and turned on, paused (issues held, no messages) or deleted (open issues close as "Closed: rule removed", after a warning).	Must
FR-2.5	Network-wide rule changes shall need two people: one school IT admin proposes, another IT admin or a leadership member approves.	Must
FR-2.6	A new check type shall be addable as one new class behind the check interface, without changing other code.	Must

Starter rules:

Rule	Check type	Applies to	Severity	Reminder
R1 School files in the school shared drive	Location	All	Normal	2 days
R2 No public links on sensitive files	Public link	Confidential, Restricted	Urgent	1 day
R3 Restricted files stay inside	Outside sharing	Restricted	Urgent	1 day
R4 Student records in Student Records	Location	Mark books, reports, behaviour, medical, special needs, contacts	Normal	2 days
R5 Only the right people open Restricted files	Internal access	Restricted	Normal	2 days
R6 No exam sharing before the exam	Sharing before date	Exams	Urgent	1 day
R7 One official copy per policy	Duplicates	Policies	Normal	3 days
R8 Leavers move school files first	Owner leaving	All	Normal	2 days
R9 No files owned by shared accounts	Generic owner	All	Normal	2 days
R10 No unapproved deletion of important files	Deleted or trashed	Restricted, exams, policies	Urgent	Same day

3.3 Issues and follow-up (FR-3)

ID	Requirement	Priority
FR-3.1	The system shall create one issue per file per broken rule, with owner, coordinator, rule version and status.	Must

ID	Requirement	Priority
FR-3.2	Open statuses: Owner notified, Reminder sent, Needs follow-up, Needs review, Owner disagrees (note required), Owner asked for help.	Must
FR-3.3	Closed statuses: Fixed, Allowed exception, Rule paused, Closed: rule removed.	Must
FR-3.4	Only a re-check shall close an issue as Fixed. "I've fixed it" re-checks that file within 2 minutes; if it still fails, the issue stays open and the owner is told what is still wrong.	Must
FR-3.5	A reminder shall be sent after the rule's reminder days; the issue moves to Needs follow-up one school day later.	Must
FR-3.6	Owner disagrees: the coordinator keeps the rule (back to Reminder sent, saved as feedback) or allows an exception.	Must
FR-3.7	Owner asked for help: the coordinator helps and marks fixed (triggers re-check), reminds, or allows an exception.	Must
FR-3.8	Exceptions need a reason and an end date; on expiry or manual end the rule applies again and the owner is told.	Must
FR-3.9	Coordinator answers on Needs review files shall be saved as labelled feedback to improve the checker.	Should

3.4 Notifications (FR-4)

ID	Requirement	Priority
FR-4.1	Each message shall state what is wrong, why it matters, how to fix it, and that Sijill changed nothing.	Must
FR-4.2	Recipients by event shall follow the matrix below.	Must

ID	Requirement	Priority
FR-4.3	Non-urgent messages shall wait for quiet hours to end (20:00–07:00 and Friday–Saturday).	Must
FR-4.4	Each person chooses message language (Arabic, English, both) and per-issue or daily summary.	Must
FR-4.5	Emails go from a dedicated sender account and every message is logged.	Must
FR-4.6	The header shall show in-app notifications with an unread count and Mark all as read.	Must

Event	Owner	Coordinator	Principal	Leadership
Normal issue, reminder, follow-up	Yes	List	Feed	–
Urgent (R2, R3)	At once	Copied	Alert	If Restricted student or health data
Exam leak (R6), important deletion (R10), 20+ deletions in an hour	Yes	Alert	Alert	Alert
Urgent still open after 1 school day	Reminder	Follow-up	Alert	Alert
Connection lost over 1 hour	–	–	Alert	Alert, IT paged
Weekly report	–	–	Own school	Network (Sunday)

3.5 Users, roles and access (FR-5)

ID	Requirement	Priority
FR-5.1	Staff sign in only with their own school account (OpenID Connect) and MFA; no Sijill passwords or sign-up.	Must

ID	Requirement	Priority
FR-5.2	Only the five school domains and head office are accepted; unknown accounts see "Your account is not set up in Sijill yet. Contact your school IT."	Must
FR-5.3	Generic or shared accounts cannot sign in.	Must
FR-5.4	Roles and sites come from directory groups; IT overrides are logged and last until the directory changes.	Must
FR-5.5	Every permission shall be enforced on the server per the permissions matrix below, never by hiding buttons only.	Must
FR-5.6	Setting a person to Leaving shall trigger R8 for their personal-drive files; Left shall remove access and move open issues to the coordinator.	Must
FR-5.7	IT can sync the directory, search and filter people, edit role, site or status, and Send welcome to new staff.	Must

Action	Teacher	Coordinator	Principal	School IT	Leadership
Own issues, preferences, help	Yes	Yes	Yes	Yes	Yes
Another teacher's files or issues	No	Own subject	Own school	Counts only	Totals only
School score and activity log	No	Own subject	Yes	Yes	Yes
All schools, incidents, weekly report	No	No	No	No	Yes
Classify unclear files, decide disputes, record follow-ups	No	Own subject	No	No	No
Allow an exception	No	Own subject	No	No	No
End an exception	No	Own subject	No	Own school	No
Run the first check	No	No	No	Own school	Yes
School-level rules	No	No	No	Own school	View

Action	Teacher	Coordinator	Principal	School IT	Leadership
Network-wide rules	No	No	No	Propose or approve	Approve
People, folders, roles, types, settings, connection	No	No	No	Own school	No
Download the activity log	No	No	Own school	Own school	All

3.6 Screens (FR-6)

ID	Requirement	Priority
FR-6.1	Teacher: My files (one card per issue with urgent badge, status, why it matters, numbered fix steps, and I've fixed it / This looks wrong / I need help), My planning, The rules, Help.	Must
FR-6.2	Coordinator: Follow-up (items needing her first; filters All open, Needs me, Urgent, Closed; search; Mark all N follow-ups as done), Planning review, My subject, People, Exceptions, Activity log.	Must
FR-6.3	Principal: My school (score badge, daily trend, four counts, issues by rule; no action buttons), Notifications, Planning review (view), Exceptions, Activity log.	Must
FR-6.4	School IT: Overview, Rules, People, Settings, Activity log.	Must
FR-6.5	Leadership: Overview (Run the first check, six tiles, schools table, Urgent right now, weekly report, accuracy card), Incidents, Exceptions, Rules (view), Activity log.	Must
FR-6.6	File details: what Sijill understood with evidence, location and sharing, each rule's status, message sent, history, and a note that Sijill only reads.	Must

ID	Requirement	Priority
FR-6.7	Every screen shall handle empty, not-yet-checked, disconnected, skipped-file, paused-rule, leaver and quiet-hours states, and confirm destructive actions.	Must
FR-6.8	A bilingual staff welcome page shall explain what Sijill does, what each role sees and what Sijill can and cannot see.	Should

3.7 History and activity log (FR-7)

ID	Requirement	Priority
FR-7.1	Per-file history: understanding, location, sharing, rules, messages, actions and changes between checks.	Must
FR-7.2	Activity log of every Sijill and user action and settings change, with before and after values.	Must
FR-7.3	The log shall be append-only and hash-chained, with a daily anchor stored in the backup vault.	Must
FR-7.4	The log can be filtered and downloaded as CSV according to role.	Must

3.8 File health score (FR-8)

ID	Requirement	Priority
FR-8.1	Per school, the system shall compute the score with the formula below, never below 0, excluding exceptions and closed issues.	Must
FR-8.2	Bands: Good 85+, Watch 70–84, Needs attention below 70.	Must
FR-8.3	The score shall be stored daily for the trend and never calculated per person.	Must

$$\text{score} = 100 - \frac{\text{open normal} + 3 \times \text{open urgent}}{\text{files checked}} \times 100$$

3.9 Settings (FR-9)

ID	Requirement	Priority
FR-9.1	Connection: status, domain, scopes, schedule, last full check, Test connection (proves a write is refused), Run a full check now, Disconnect (red banner on every page) and Reconnect.	Must
FR-9.2	Sites and folders: rename, add (shows "Waiting for first check"), edit or remove a site (history kept; head office cannot be removed); manage standard folders.	Should
FR-9.3	Roles: rename, see permissions, directory group and member count.	Should
FR-9.4	14 document types with default sensitivity changeable in one click; a new type needs labelled examples.	Should
FR-9.5	Messages: language, delivery, quiet hours, sender address, reminder defaults.	Must
FR-9.6	Privacy and data: log retention (1, 2 or 5 years), AI service, masking always on, log download.	Must

3.10 Planning-file review (FR-10)

ID	Requirement	Priority
FR-10.1	A teacher submits subject, grade, week, type and a drive link (no file copy).	Must
FR-10.2	The coordinator approves or returns with a reason; resubmissions keep history.	Must
FR-10.3	Statuses: Not submitted, Submitted, Returned, Approved; one reminder on the due day.	Must

ID	Requirement	Priority
FR-10.4	Teachers see their own, coordinators their subject, principals totals.	Must

3.11 Exports and reports (FR-11)

ID	Requirement	Priority
FR-11.1	All downloads (CSV, PDF) shall use initials or IDs; full names appear only on screen to permitted roles.	Must
FR-11.2	IT can unmask one export for a logged reason.	Should
FR-11.3	Every exported document footer shows who, when and sensitivity.	Must
FR-11.4	The weekly report (PDF) is produced every Sunday per school and for the network.	Must
FR-11.5	Adoption measures per school (never per person): weekly active staff, time to fix, fixed without follow-up, plans on time.	Should

3.12 Backup and deletion protection (FR-12)

ID	Requirement	Priority
FR-12.1	New and changed files shall be copied hourly to the backup vault, with nightly reconciliation of all files.	Must
FR-12.2	Native documents shall be exported to DOCX, XLSX, PPTX or PDF; each run writes a manifest (path, drive, owner, sharing, checksum).	Must
FR-12.3	Each copy shall be verified by checksum; failure alerts School IT.	Must

ID	Requirement	Priority
FR-12.4	Coordinators or principals request a restore from file history; School IT approves (two people for a whole drive or school) and restores with workspace tools.	Must
FR-12.5	The system shall flag nightly any teacher with more than contributor rights on a subject drive (drift).	Should
FR-12.6	One random file per school shall be test-restored monthly and one subject drive per term; results shown in Settings › Backup.	Must

4. External interface requirements

Sijill has one web user interface and four outside connections; every connection sits behind its own interface so a provider can be swapped without changing the rest.

4.1 User interface

- Arabic first, right to left, Arabic-Indic digits; English optional with a language switch; all text in resource files.
- Responsive for phone, tablet and laptop; WCAG 2.1 AA.
- Sign-in page: one "Continue with school account" button, a line that only network accounts work, and a link to the welcome page.
- Header: notifications with unread count, user menu (My preferences, Help, Sign out).
- Wording: buttons say what happens ("I've fixed it", not "Resolve"); staff never see "violation", "non-compliant", "escalated" or "audit failure".
- Errors in plain words without internal detail; every destructive action asks for confirmation.

4.2 Software interfaces

Interface	Direction	Purpose	Contract
Workspace connector	Read only	List changes, get file, get permissions, get people, audit feed	`WorkspaceConnector` interface; one read-only service account per school domain
AI service	Out (masked text), in (profile)	Classify document type, sensitivity, evidence	`AiClient` interface; in-country model, no training, zero retention, structured output
Email sender	Out	Issue messages, reminders, alerts, weekly report	Dedicated sender account; every send logged with delivery status
Backup vault (Provider B)	Out	Encrypted file copies, manifests, database copies, log anchors	Write-once, versioned storage; separate account and keys
Identity provider	In	Staff sign-in and MFA	OpenID Connect via the school workspace

4.3 Communication interfaces

- All traffic uses TLS 1.3; HTTPS only, HSTS on.
- The database has no public endpoint; services talk over a private network.
- File readers run in isolated, network-restricted workers.
- Checks, messages and backups run as queued jobs that resume after an outage.

5. Non-functional requirements

Sijill is judged first on accuracy and trust: zero writes to school files and zero Restricted files reported as fine.

5.1 Performance, scale and availability

ID	Requirement	Target
NFR-1	Nightly full check	60,000 files in under 4 hours
NFR-2	Single-file re-check ("I've fixed it")	Under 2 minutes
NFR-3	Change detection from the feed	Within 10 minutes
NFR-4	Screen load	Under 2 seconds
NFR-5	Scale without redesign	50 sites, 5,000 staff, 1 million files (horizontal workers)
NFR-6	Availability	99.5% in school hours (07:00–16:00, Sunday–Thursday)
NFR-7	Resilience	Queued work resumes after any outage; nothing lost

5.2 Checker quality

Measured on a held-back 20% of labelled synthetic files never used for tuning, against a keyword-only baseline. Targets change only by written agreement after the first baseline run.

ID	Measure	Target
NFR-8	Writes to school files	0 (failing write test on every release)
NFR-9	Restricted files reported fine	0
NFR-10	Recall on Confidential and Restricted	95% or more
NFR-11	Precision	85% or more
NFR-12	Document type accuracy	90% or more
NFR-13	Arabic and mixed-language accuracy	Within 3 points of English
NFR-14	Real leadership alerts (not false alarms)	95% or more

5.3 Usability, maintainability and observability

- **NFR-15 Language:** Arabic first, right to left; English optional; languages as resource files.
- **NFR-16 Accessibility:** WCAG 2.1 AA on every screen.
- **NFR-17 Configuration over code:** rules, roles, sites, document types, languages and timings editable in the app.
- **NFR-18 Maintainability:** one connector interface and one AI interface; automated tests on every change; a clear README.
- **NFR-19 Observability:** IT health dashboard for checks, skipped files, queue, backup and message delivery.
- **NFR-20 Usage limits:** AI and message usage metered per school, limits set by IT, alert near the limit.

6. Security and privacy requirements

A compromise of Sijill must not be able to change, share or delete any school file, or leak readable student data.

6.1 Threats and required controls

ID	Threat	Required control
SEC-1	Stolen service credentials	Read-only scopes per school; failing write test each release; keys in a vault, rotated every 90 days
SEC-2	Staff account takeover	SSO with MFA for all; phishing-resistant keys for IT and Leadership; new-device alerts; sessions 8 hours (30 minutes idle for IT and Leadership)
SEC-3	Shared accounts	Generic accounts cannot sign in; R9; every action tied to one person
SEC-4	Cross-school access (e.g. URL editing)	Server-side permission check on every request; row-level security by site and subject; access tests per role

ID	Threat	Required control
SEC-5	Exam leak	R6 detection within minutes; leadership alert; exceptions need reason and end date
SEC-6	Important deletion or ransomware	R10 and bulk-deletion alerts; write-once backups at a separate provider; restores tested each term
SEC-7	Student data sent to AI	Masking before every call; hard checks; in-country AI, no training, zero retention
SEC-8	Log tampering	Append-only hash chain; daily anchor in the vault; database role cannot update or delete log rows
SEC-9	Malicious files	Isolated, network-restricted readers; size and type limits; locked or oversized files skipped and logged
SEC-10	Web attacks	Parameterised queries, output encoding, strict CSP, CSRF tokens, secure cookies, rate limits, WAF
SEC-11	IT insider misuse	Masked evidence; counts not content; logged views of Restricted details; two-person approval for network rules
SEC-12	Data leaving the country	Every component in-country; a data-location register

6.2 Controls by layer

- **Data:** TLS 1.3 in transit; AES-256 at rest; keys in an in-country key store; every table carries `org_id` and `site_id` under row-level security.
- **Infrastructure:** private network, no public database endpoint, least-privilege identities, hardened and signed images.
- **Monitoring:** alerts to School IT for failed sign-ins, permission errors, unusual exports and connector failures.
- **Supply chain:** pinned dependencies, vulnerability and secrets scanning, code review on every change.

- **Assurance:** senior code and security review and an independent penetration test before go-live and yearly; quarterly access review by each principal.

6.3 Privacy

- Comply with Qatar's Personal Data Privacy Protection Law (Law No. 13 of 2016) and the network's data policy.
- Store metadata, sharing, masked evidence, profiles, issues, messages and logs, never file content.
- Logs never hold student names, IDs or health details.
- Student records, health, special-needs, behaviour and contact files default to Restricted.
- Retention: activity log 2 years by default (1, 2 or 5 configurable); issues and profiles for the file's life plus 1 year.
- School IT can export or delete a person's records on request; the deletion is logged.
- Incident playbook: contain, assess, notify (principal, leadership, regulator if required), recover, review.

7. Data requirements and backup

One relational database holds every record; every screen is a view of it, and file copies exist only in the backup vault.

7.1 Main entities (Phase 1)

Group	Entities	Notes
Files	File, FileSnapshot, FileProfile	Snapshot holds location and sharing per check (grantees incl. anyone and external) as JSON
Rules and issues	Rule, RuleVersion, Issue, IssueEvent, RuleException, Message, Incident	Message covers email and in-app via a channel field
People and audit	Person, ActivityLog	Person holds directory data, overrides, status (new, active, leaving, left) and preferences; ActivityLog is append-only and hash-chained

Group	Entities	Notes
Planning and restore	PlanningSubmission, RestoreRequest	Review decision and reason stored on the submission; history in the log
Configuration	Org, Site, Subject, Folder, Role, DocType, Setting, JobRun, SchoolScore	Role holds its directory group; JobRun records check and backup runs; SchoolScore is stored daily

Example file profile:

```
{
  "doc_type": "Mark book",
  "site": "School B",
  "subject": "Mathematics",
  "has_student_data": true,
  "sensitivity": "Restricted",
  "confidence": 0.96,
  "evidence": ["A table of 28 [NAME] entries next to term marks", "A column
of [STUDENT_ID] values"]
}
```

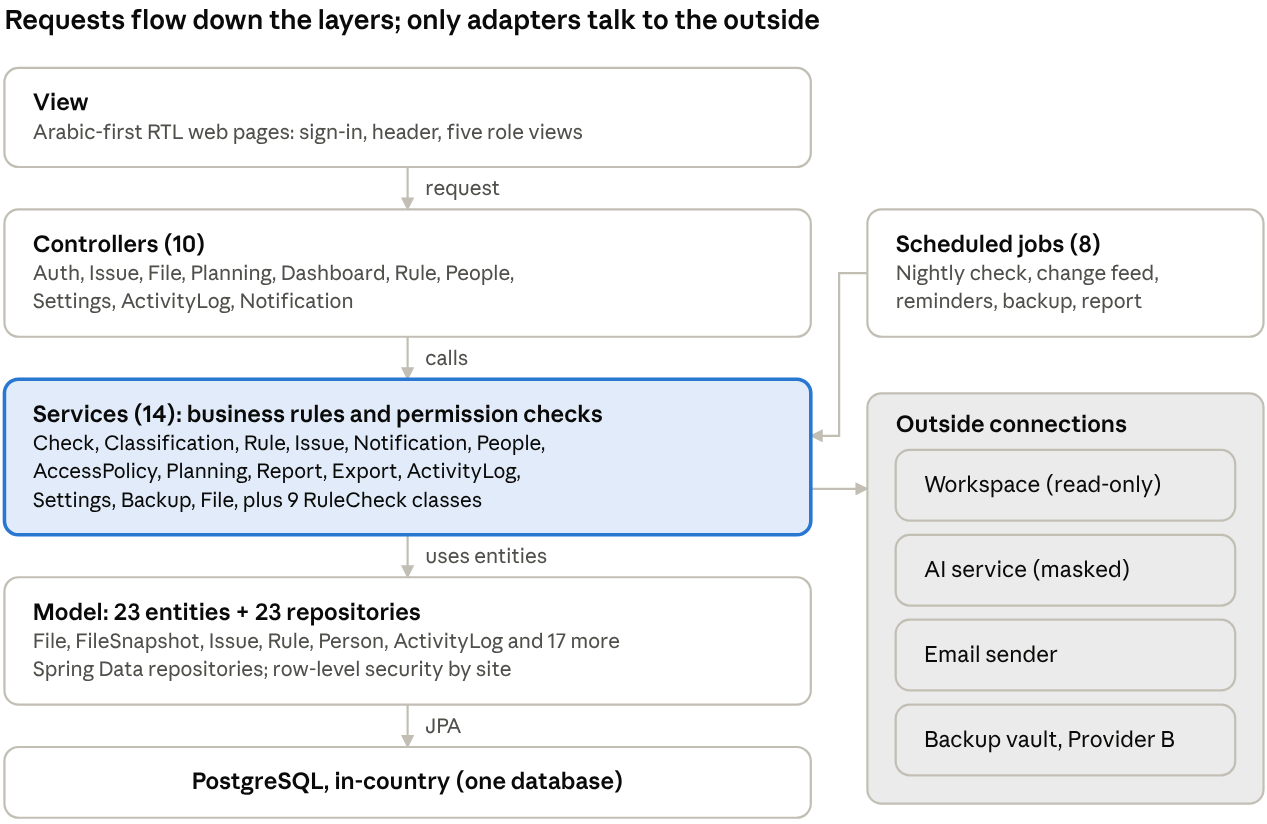
7.2 Backup and recovery targets

ID	Item	Target
DR-1	Recovery point, files	1 hour (changed files); nightly reconciliation
DR-2	Recovery point, database	15 minutes (point-in-time recovery)
DR-3	Recovery time	One file same school day; a subject drive 1 school day; a whole school 3 school days; Sijill itself 4 hours
DR-4	Copies	3-2-1: three copies, two providers, one immutable; no copy or key leaves the country

ID	Item	Target
DR-5	Vault retention (immutable for the same period)	Hourly 7 days, daily 30 days, monthly 12 months, end-of-year 5 years or records policy if longer
DR-6	Database copies	35 daily and 12 monthly
DR-7	Log anchors	5 years
DR-8	Plan B	Rebuild Sijill in a second in-country region from code and the latest vault copy within 4 hours

8. System design (MVC)

Sijill uses a lean Spring MVC design of 47 core classes: 10 controllers, 14 services and 23 model entities. A controller only handles the request and picks the view; every business rule and permission check lives in a service; entities and repositories hold the data.



Sijill MVC layers · 10 controllers, 14 services, 23 entities

Controllers never call the database or outside systems directly. Scheduled jobs call the same services as the screens, so a nightly check and an "I've fixed it" re-check run the same code.

8.1 Controllers

Controller	Screens and actions	Used by
AuthController	Sign-in, sign-out, unknown-account page	All
IssueController	My files, issue actions (I've fixed it, This looks wrong, I need help), Follow-up, Exceptions, message screen	Teacher, Coordinator, Principal
FileController	File details, file history, restore request	All (scoped)

Controller	Screens and actions	Used by
PlanningController	My planning, submit, Planning review, approve or return	Teacher, Coordinator, Principal
DashboardController	My school, Overview, Incidents, weekly report, Run the first check	Principal, Leadership, School IT
RuleController	Rule list, editor, impact and message preview, pause, delete, propose and approve	School IT, Leadership
PeopleController	Directory sync, search, edit role, site or status, Send welcome, mark followed up	School IT, Coordinator
SettingsController	Connection, sites, folders, roles, document types, messages, privacy, backup status	School IT
ActivityLogController	Log view, filters, CSV download, name-free exports	Coordinator, Principal, School IT, Leadership
NotificationController	Header notifications, mark all read, My preferences, Help, welcome page	All

8.2 Services

Service	Responsibility
CheckService	Runs full checks and the change feed, reads files through the connector, applies active rules
ClassificationService	Text extraction and OCR, masking, hard checks, AI profile, confidence routing to Needs review
RuleService	Rule versions, dry-run impact preview, pause and delete, two-person approval
IssueService	Issue lifecycle, re-check, status changes, exceptions, incidents

Service	Responsibility
NotificationService	Decides who is told, writes bilingual messages, quiet hours, sends and logs
PeopleService	Directory sync, overrides, leavers, preferences
AccessPolicyService	Server-side permission check for every request, by role, site and subject
PlanningService	Planning submissions, reviews, due-day reminders
ReportService	School health score, weekly report PDF, adoption measures
ExportService	Name-free CSV and PDF exports, logged unmasking
ActivityLogService	Append-only, hash-chained log and daily anchor
SettingsService	Sites, folders, roles, document types, settings, connection test
BackupService	Hourly and nightly copies, manifests, verification, restores, deletion-drift check
FileService	File details, per-file history, snapshots

Supporting classes outside the three layers:

- `RuleCheck` interface and 9 check classes, one per check type (FR-2.6).
- `WorkspaceConnector` interface and one implementation per workspace platform.
- `AiClient` interface and one implementation for the in-country model.
- 8 scheduled jobs: nightly check, change feed, reminders, hourly backup, nightly reconciliation, exception expiry, weekly report, deletion drift.
- Security configuration (OpenID Connect, MFA, session rules) and about 8 enums (Sensitivity, Severity, IssueStatus, CheckType, PersonStatus, RoleType, PlanningStatus, Language).

8.3 Example request: "I've fixed it"

1. `IssueController` receives the button press from the teacher's My files page.

2. `AccessPolicyService` confirms the issue belongs to this teacher; otherwise the request is refused.
3. `IssueService` asks `CheckService` to re-check that one file through the read-only connector.
4. `CheckService` runs the issue's `RuleCheck` on the fresh snapshot.
5. Pass: `IssueService` closes the issue as Fixed. Fail: it stays open and `NotificationService` tells the owner what is still wrong.
6. `ActivityLogService` records the action; the controller returns the updated card to the view.

8.4 Package structure

```
qa.sijill
├── controller   (10 controllers)
├── service      (14 services)
├── model        (23 entities + enums)
├── repository   (23 Spring Data repositories)
├── check        (RuleCheck + 9 check classes)
├── connector    (WorkspaceConnector + implementation)
├── ai           (AiClient + implementation)
├── job          (8 scheduled jobs)
└── config       (security, scheduling, queue)
```

9. Acceptance criteria and release gates

Phase 1 is accepted when all eight stage gates pass, all Must requirements are demonstrated, and no high or critical security findings are open. Progress is measured by gates passed, not dates.

1. **Foundations:** test domains, one read-only account each, repository, database, containers, tests, README.
   - Gate: one script lists every file and its sharing; a write attempt fails.
2. **Synthetic world:** about 120 staff, 1,500 student records, 400–600 Arabic, English and mixed documents incl. scans, seeded mess and about 10% tricky cases, all labelled.
   - Gate: realistic and resets in one command.
3. **Understanding and evaluation:** reader, Arabic OCR, masking, hard checks, AI profile, keyword baseline, evaluation report.

    - Gate: held-back results meet section 5.2.
4. **Rules and issues:** all 9 check types, rules as data, statuses, re-checks.
    - Gate: issues match labels; zero Restricted files reported fine.
5. **Messages, follow-up, incidents:** bilingual messages, reminders, quiet hours, exceptions, disputes, help, leadership alerts.
    - Gate: a simulated school week is correct and nothing is lost.
6. **Backup vault:** hourly and nightly copies, manifests, immutability, verification, restore.
    - Gate: a deleted subject drive is restored within 1 school day.
7. **The app:** sign-in and the five role views, then Rules, People, Activity log, Settings, notifications, preferences.
    - Gate: each role does its job unaided; permissions enforced on the server.
8. **Security and launch:** code and security review, penetration test, welcome page, rehearsed demo.
    - Gate: no open high or critical findings; the demo runs twice from a reset.

If time runs short, Sites, Roles and Document types settings (FR-9.2–FR-9.4) can come last.

9.1 Test approach

- **Unit and integration tests:** JUnit and Testcontainers on every change; one test per check type against labelled files.
- **Write test:** every release attempts a write with the service account and must fail.
- **Access tests:** for each role, every endpoint is called with another school's and another subject's IDs and must be refused.
- **Week simulator:** plays a school week (reminders, follow-ups, fixes, exceptions ending) and checks every issue ends fixed, as an exception, or visibly open.
- **Restore tests:** monthly file restore and termly subject-drive restore, results recorded.

Appendix: Phases 2–4 summary

Later phases build on the same database, log, vault and controls; each starts only after the previous phase passes its gates, and each will get its own SRS.

Phase	Scope	Main requirements	New roles
2 — School portal	Separate "worlds" for staff, parents and students	Registration and bulk family accounts (PT-01–PT-06); lessons, weekly plan, revision library (PT-07–PT-11); homework and marking (PT-12–PT-15); messages and notices (PT-16–PT-21); year-long points and top five (PT-22–PT-27); dashboards with masked exports (PT-28–PT-31)	Parent, Student, Registrar
3 — AI for teachers	Optional tools grounded in Ministry-approved lessons	Lesson-plan generator, differentiation, competency check, question generator, local wrong-question filter (AI-01–AI-13)	None
4 — AI for students and families	Beta in the pilot school first	Textbook-only chatbot, off-topic refusal, page-and-hints first, homework helper (AI-14–AI-20)	None
Later	To be scoped	Mobile apps, photo feed with consent, timetable links, counsellor role, QES data link, other curricula (PT-32–PT-36)	Counsellor

AI quality targets for Phases 3–4: grounded, cited answers 100%; off-topic correctly refused 95%+; plans usable without major edits 80%+; Arabic answers rated correct and natural 90%+.